A Review of Machine Learning Techniques for Predictive Maintenance in Industry 4.0

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***Abstract*—Decision-making requiring a large amount of data input and customization in the industrial process confronts both equipment and management on a daily basis. One of the primary issues in this domain is the capacity to predict when equipment will need maintenance. Predictive maintenance is an option that improves production quality and reduces machine downtime. Our observation is that assessments and seminars on Industry 4.0 (I4.0) mostly concentrate on data analytics and machine learning techniques to transform production processes, neglecting the implementation of predictive maintenance approaches. This study provides a review of machine learning based predictive maintenance activities in I4.0. It aims to identify and categorize various approaches and applications related to this field. The primary contribution of this review is the examination of the existing constraints in predictive maintenance, taking into account the requirements of I4.0.**

Keywords— Machine learning, Industry 4.0, Predictive maintenance, Deep learning, Industrial automation

I. Introduction

Industry 4.0 (I4.0) is the fourth industrial revolution that occasioned by the swift transformation of the industrial world brought about by the advancement of technologies that connect the digital and physical realms [1], [2]. These technologies include robotics and cutting-edge devices, the internet of things, and artificial intelligence's contribution to smart automation [3], [4], [5]. The demonstrated scalability and automation of machine learning's predictive power hold great potential in this context, given the scope of industrial problems [1]. As a result, in an effort to reap these advantages, businesses and researchers have been investigating machine learning for I4.0 more and more in recent years.

The field of industrial applications of "machine learning" (ML) is complex because of the many interconnected areas. The current surge in interest, albeit beneficial, has made it difficult to comprehensively track the developments within published research. An unambiguous perspective of the area on a large scale enables interested parties to observe unmistakable indications of a swift growth in interest and, more precisely, identify the areas that capture the community's attention [6]. This piece of information is crucial as it enables one to deduce important aspects such as the study direction that can provide valuable contributions or the practical and worthwhile solution directions [7].

In current generation of information, data plays a crucial role in enabling the anticipation or collaboration in making predictive decisions [1]. Predictive Maintenance Decision (PMD) refers to the maintenance decision that is based on systematic use of historical data, models, and domain expertise [8], [9]. By using statistical or machine learning models, it is possible to forecast trends, behaviour patterns, and correlations in order to anticipate future failures [10]. Existing definitions of intelligent industry or intelligent manufacturing are derived from the deployment of PMD and approaches designed to enhance manufacturing [6]. The maintenance has become crucial for companies because of the rising complexity of the linkages among various production activities in widely expanded manufacturing ecosystems [11].

The "Internet of Things" (IoT) is another notion that enhances the value of the PMD process. The specialization known as Industrial IoT involves the application of IoT technologies in an industrial setting, integrating Machine Learning and Big Data [11]. Proficiency in the selection of Machine learning algorithms that are appropriate for a particular task or problem is a fundamental skill for data analysts and practitioners seeking to implement machine learning in industrial environments [12]. This skill can determine whether failure or success is achieved. nevertheless, without prior experience or prior research of a similar kind, it is difficult to imagine a method to implement machine learning at the industrial level to enhance business performance [9]. This is particularly true given the large number of algorithms that have been suggested in related literature.

The level of diversity PMD can be confusing and deceptive, and the issue is exacerbated by the absence of a comprehensive collection of optimal applications for every

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sector and company [9]. We propose that a comprehensive literature review, specifically examining the various techniques of machine learning for predictive maintenance in industrial domain, could be highly beneficial in identifying current and future trends [8]. Moreover, it can guide industrial practitioners in the careful selection and deliberate use of ML techniques.

An analysis of machine learning-based predictive maintenance activities in I4.0 is carried out in this study [13], [14]. The objective is to systematically identify and classify different methodologies and implementations associated with I4.0. The remaining sections of the paper are organised as follows. Section 2 provides a concise introduction of predictive maintenance decision. Section 3 provides the existing constraints in I4.0. Section 4 outlines the various machine-learning techniques that are compatible with I4.0. Section 5 concludes the study with recommendation for future research.

II. Predictive Maintenance Decision

Predictive maintenance decision (PMD), a notable multidisciplinary research group suggests creating and integrating research lines in data gathering, infrastructure, storage, dissemination, security, and intelligence.. The analysis of data from several sensors in Industry 4.0 (I4.0) settings opens up new options for determining an asset's remaining lifespan [15], [16]. The idea that PMD can create scheduling action by taking into account equipment performance or conditions over time is both exciting and crucial to the industry's future [17]. Making effective predictive maintenance decisions requires a substantial quantity of data from every step of the manufacturing process. Consequently, it can save maintenance expenses and downtime while simultaneously raising I4.0 production and quality.

Forecasting an equipment residual span is a natural challenge in engineering, mechanics, and automation applications [17], [18]. Within this context, PMD research is directly related to the observation axis of prognostics and systems health management, and the use of intelligent tools to predict failure phenomena [19]. Maintenance falls into four categories of occurrence: “corrective”, “preventive”, “predictive”, and “prescriptive”. Maintenance in the corrective scenario occurs either when the defect is detected or when indicative signs are present. Preventative maintenance adheres to set schedules. In contrast, PMD uses time-based data and knowledge to quickly report a potential breakdown, avoiding any periods of idleness. Prescriptive maintenance is about how to regulate the incidence of a particular event [15].

The prognostics and systems health management, divides maintenance procedures into four categories: “corrective”, “fixed-interval preventative”, “failure detection”, and “Condition-Based” Maintenance. Asset monitoring in this context employs sensors, algorithms, and mathematics to determine the residual span of equipment in I4.0.

III. Challenges in Industry 4.0

A. Equipment Residual Span Prediction

Predicting the remaining useful time of an equipment can improve efficiency and planning in industry maintenance. This is applicable to parts like batteries, bearings and complex processes like ion milling that need to be changed on a regular basis. Sequence models are commonly used to address the temporal aspect of this problem, but creative approaches can also be used [6], [10], [20].

B. Anomaly Detection

Effective monitoring of machine anomaly is crucial in the industrial sector, as it impacts safety, efficiency, and costs. Scalable automation of fault detection systems using machine-learning approaches is a popular application in the field [10], [20], [21]. Visual inspection is a crucial part of fault detection systems, as it detects degradation or surface problems in manufactured items. As a result, convolutional neural networks are commonly used in systems that aim to automate its processes. Despite the feat and acceptance of convolutional neural networks, some continue to apply them to time-series data through visualisations.

C. Load Forecasting

Load forecasting involves projecting variations in request over time, especially for power systems. Accurate load forecasting is crucial for meeting rising demands[6]. Consider a cassava process firm that may require supplementary resources, such as fuel to meet anticipated demand. Accurate forecasting can increase both fiscal and ecological efficiency in energy usage. In I4.0 application case, deep learning models such as recurrent neural networks or long short-term memory outperform other models for temporal regression [6], [10], [20]. Although research suggests their effectiveness, they face similar hurdles in acceptance and hands-on application, such as data accessibility and sophisticated system integration.

D. Independent Operations

Leveraging machine learning for automation of repetitive tasks is a key aspect of I4.0. Robotic operation typically use convolutional neural networks, reinforcement learning, or a combination of both [20]. Reinforcement learning is good for developing dynamic agents that adapt to varied contexts, while convolutional neural networks can recognise visual elements for tasks like object detection or assisting reinforcement learning agents in observing their surroundings [6], [10], [20]. Automated pick and place and coastal crane automation for shipping containers are examples of such applications. Autonomous operations extend beyond manufacturing to include customer service and marketing. Automation in digital domains is easier and more effective due to their intrinsic nature.

E. Energy Optimization

Optimising energy use in industrial settings is similar to the load forecasting challenge discussed before. ML models are commonly used to anticipate energy consumption. Forecasts can help optimise responses to demand. In [6], IoT-based data collection helped a soap factory optimise energy consumption by giving analysis results to enhance decision-making [6], [10], [20]. Examples of energy consumption optimization include offloading machine learning computation to the edge or fog, using reinforcement learning to optimise raw materials trajectory in food and beverage factory.

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IV. Industry 4.0 Compatible Machine Learning Techniques

A. Artificial Neural Network

Artificial Neural Networks (ANN) are complex networks modelled after the simpler biological structure of neurons observed in animal brains. ANNs typically have three layers: input, hidden, and output. An artificial neural network (ANN) is represented by a set of connections between the input and hidden layers, each having its own weights and biases [22]. Artificial neurons, which are interconnected nodes in an ANN, replicate the neurons found in the biological brain. The interconnected nodes provide the same purpose as synapses in biological neurons: they transmit impulses. Figure 1 depicts the standard structure of an ANN. Equation 1 is the output of ANN, which takes x1, x2, and x3 as inputs [22].

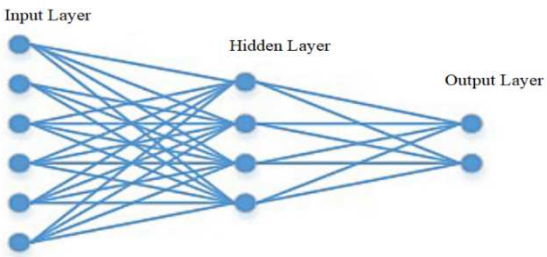

Figure 1: A typical structure of ANN [22].

$$y = A(W^T x) = A\left(\sum_{i=1} Wi\, xi + b\right) \quad (1)$$

where A is a function of activation, W is matrix weight, b is a scalar (bias).

In a typical deep learning model, the "W" parameters reflect the weights used to connect distinct neurons in the network. Equation 2 represents the mapping equation for a neuron 'i' in the kth layer [22].

$$Z_{ki} = f_{ki}\,(W_{ki} \cdot z_{k-1} + b_{ki}) \quad (2)$$

Where f_{ki} represents the neuron activation function, z_{k-1} represents the values of the neurons in the preceding layer, W_{ki} represents the matrix of weights that build connections between the neurons and b_{ki} represents the bias value of the neuron.

B. Convolutional Neural Network

One accepted and efficient deep learning architecture is the convolutional neural network (CNN). In computer vision, this neural network-based approach is frequently used for pattern recognition and classification in images. Convolutional neural network architecture typically consists of three types of layers: convolutional, pooling, and fully connected [6], [10], [20]. A convolutional layer uses a set of learnable parameters known as a kernel or filter to compute the scalar product of a tiny portion of the input image or matrix. The majority of CNN's computing cost is accounted for by these computations. Before going on to the next layer, the rectified linear unit function is used to activate the output. A pooling layer down samples by replacing some output with a statistic derived from nearby information [6], [10], [20].

C. Support Vector Machine

The Support Vector Machine (SVM) is a popular machine learning technique that uses unsupervised learning and modelling to categorise or forecast discrete or continuous information [22]. An SVM learning method creates a "non-probabilistic binary linear classifier" by assigning training data to one or more predefined categories. Figure 2 depicts how SVMs work in a two-dimensional plane. The three courses are separated by three lines. SVM is an excellent solution for detecting defects in the power segment because to its multiple major benefits [6], [10], [20].

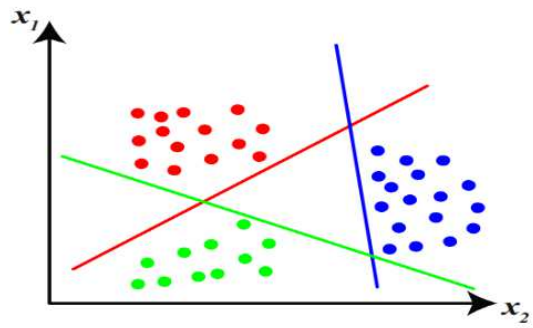

Figure 2: SVM representation in a two-dimensional space [22].

D. Decision Trees

Graphs with nodes branching off based on thresholds are called decision trees. Recursively analysing nodes or features can yield the highest predictive features [22].. A random forest is an aggregated set of decision trees that can improve performance and reduce overfitting when used to make predictions. This can be accomplished by using random forests, an ensemble learning technique, which trains trees on subsets of data or features and aggregates their output. Bootstrap aggregating, sometimes known as "bagging," is a technique for training trees on different samples or subsets of data [6], [10].

E. Reinforcement Learning

Reinforcement Learning (RL) differs from supervised and unsupervised learning approaches by rewarding and punishing conduct [6]. This approach involves allowing models to discover optimal behaviours rather than deliberately training them using large data. In RL, the model is an agent that selects actions from a predefined set of options. The agent makes decisions based on observations from its environment. [6], [10], The agent learns desired behaviour by being rewarded or penalised based on the action they choose. The training involves defining concepts like policy, reward, and value function. A policy is a function that transfers the agent's current state to an action, which might be deterministic or stochastic [6], [10], [20].

A value function calculates the expected return or reward for a specific condition based on a policy function. This allows the agent to compare various policies in a given context [6]. Deep RL occurs when neural networks are utilised to approximate previously mentioned functions. RL has successful applications in gaming, robotics, banking, recommender systems, and load allocation in telecommunications and energy grids [6], [10], [20].

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F. Generative Adversarial Networks

Generative Adversarial Networks (GANs) are unsupervised models that identify patterns in input data and generate new output samples that resemble the original set [6], The GAN architecture includes a generator for creating new samples and a discriminator for distinguishing between fake and real samples. The discriminator receives feedback based on known labels for real samples, whereas the generator receives feedback depending on the discriminator's output accuracy [6], [10]. Despite being a relatively new technology, it is quickly gaining popularity due to its capacity to generate valuable data points for complex issues with less data. In addition to training data synthesis, direct applications include image processing, such as restoration or super resolution, image-to-image translation, music production, and drug discovery [20], [22].

V. Conclusion

Maintenance involves strategic planning to ensure quality, safety, and productivity. Industry 4.0's evolving concepts present both opportunity and problems. Our study focusses on predictive maintenance decision, exploring existing approaches and applications. A survey of the literature on predictive maintenance decisions in the context of Industry 4.0 is presented in this article. Building adaptive multi-model algorithms and architecture is essential for Industry 4.0 applications, given the huge potential of machine and deep learning. Further study can also be directed towards the development of generic anomaly detection system capable of operating efficiently in industrial applications. It is also worthwhile to examine algorithms that can function effectively and efficiently on hardware with limited processing power.

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